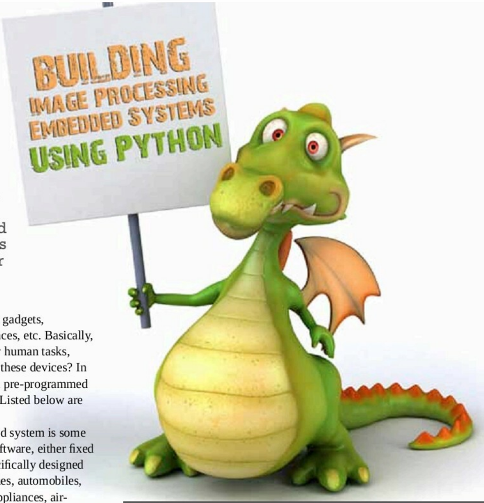

The first part of this article gives a brief overview of the embedded vision and the various components required to make it work. It also covers the installation procedure for the OpenCV library.

Part—1

Modern life is incomplete without gadgets, smartphones, automated appliances, etc. Basically, electronic devices perform daily human tasks, making life much easier. So what controls these devices? In layman's language, it's a small circuit with pre-programmed human logic, called an embedded system. Listed below are some useful definitions.

Embedded system (ES): An embedded system is some combination of computer hardware and software, either fixed in capability or programmable, that is specifically designed for a particular function. Industrial machines, automobiles, medical equipment, cameras, household appliances, air-planes, vending machines and toys (as well as the more obvious cellular phone and PDA) are among the myriad possible hosts of an embedded system.

Image processing: In electrical engineering and computer science, image processing is any form of signal processing for which the input is an image, such as a photograph or video frame. The output of image processing may be either an image, or a set of characteristics or parameters related to the image. Most image-processing techniques involve treating the image as a two-dimensional signal, and applying standard signal-processing techniques to it. In other words, it is basically the transformation of data from a still or video camera into either a decision or a new representation. All such transformations are done to achieve some particular goal. The input data may be a live video feed, the decision may be that a face has been detected, and a new representation may be conversion of a colour image into a grey-scale image.

Embedded vision: Embedded vision is the merging of two technologies—embedded systems and image-processing/computer vision (also sometimes referred to as *machine vision*). Due to the emergence of very powerful, low-cost and energy-efficient processors, it has become possible to incorporate vision capabilities into a wide range of embedded systems. One successful example is the Microsoft Kinect video game controller, which uses embedded vision to track the movements of Xbox 360 users, avoiding the need for handheld controllers. Microsoft sold eight million Kinect units in the first two months after its introduction.

Moving on, the five basic components required to build an embedded system using Python are discussed below, in brief.

The operating system (OS)

The OS is the heart of an embedded vision system. Many different dependencies that are required to run software are provided by the OS, which is the interface between the hardware and the software. Of the many OSs in the market (such as Windows, Linux, etc) there are reasons why Linux is preferable:

- It is open source, i.e., it is free of cost and free, as in 'freedom'.
- It is virtually virus-free.
- It has pretty good support—bug trackers, documentation, forums, mailing lists, IRC, etc.
- It is easy on systems resources, and more resources can be dedicated to applications, which means faster processing.

There are various Linux distributions like Ubuntu, Fedora, etc, to choose from. I personally prefer Ubuntu, because of its easy learning curve. Now, size is the main constraint for an embedded system. Using a PC OS won't do. A much more portable solution is required, such as the PandaBoard shown in Figure 1. Check out its features: